Your Paper Title Here

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Abstract

Write your abstract here.

Keywords

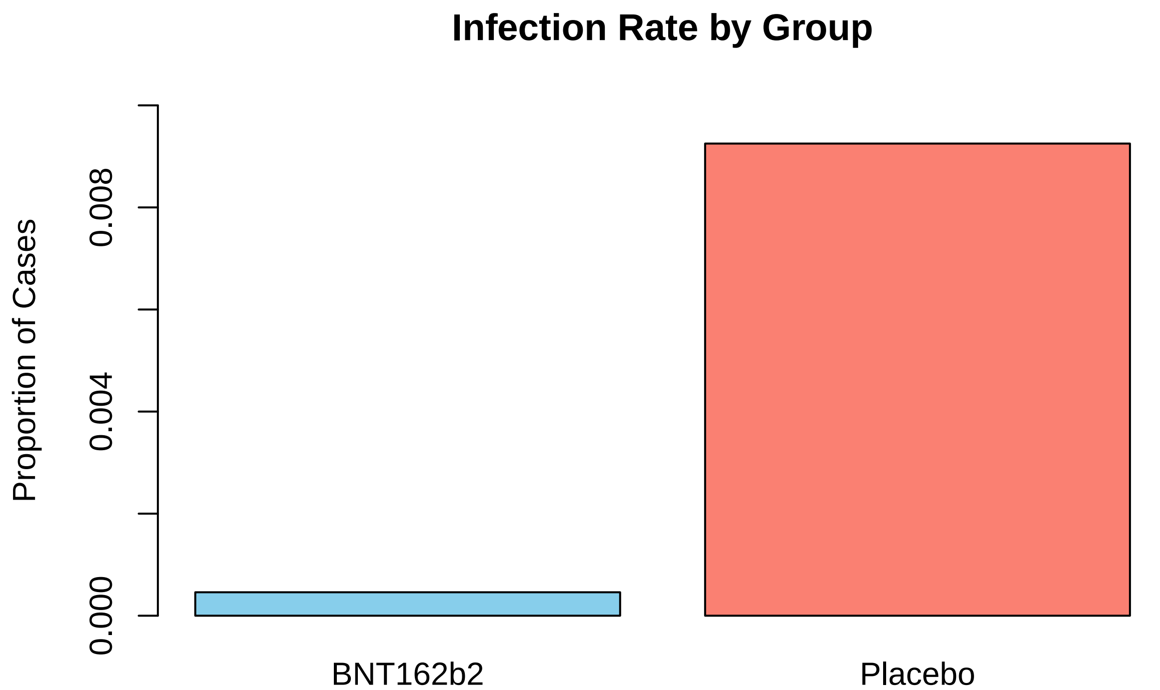
Keyword 1, Keyword 2, Keyword 3, Keyword 4

Introduction / Background

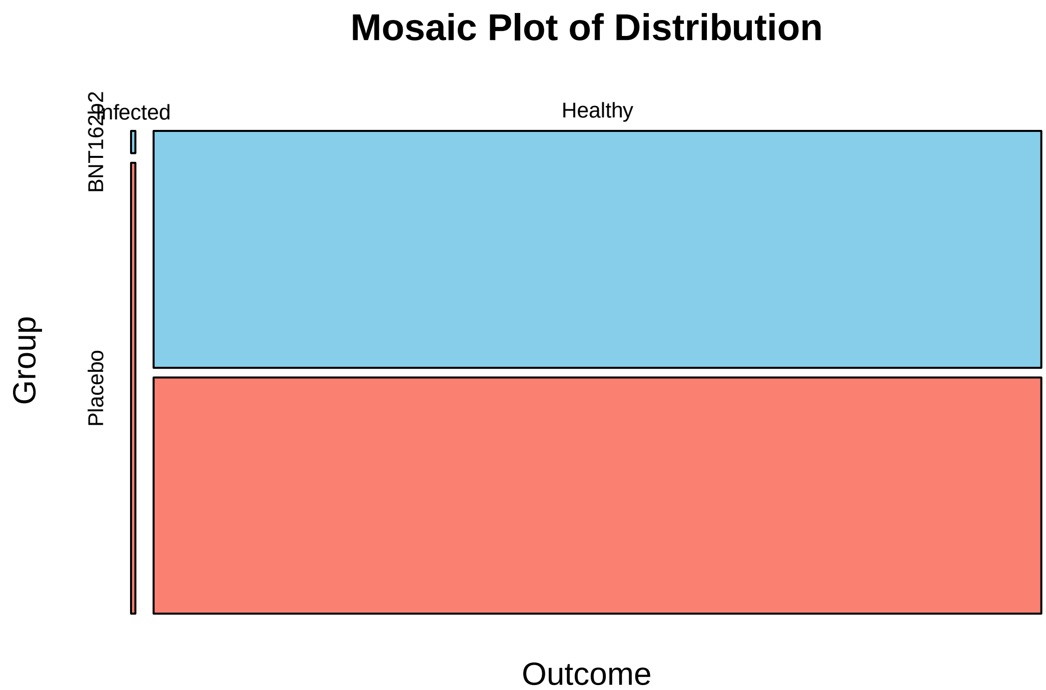
The December 2020 Emergency Use Authorization (EUA) of the BNT162b2 vaccine marked a pivotal shift in the global effort to mitigate the SARS-CoV-2 pandemic. Developed through a collaboration between Pfizer and BioNTech, this two-dose regimen represented a departure from traditional vaccine technologies. As noted in contemporary scientific literature, while viral vector approaches were utilized by manufacturers like Janssen and AstraZeneca, BNT162b2 utilized lipid nanoparticle-formulated nucleoside-modified RNA (mRNA) to induce an immune response against the viral spike protein. To evaluate the safety and clinical impact of this technology, a rigorous, multinational, placebo-controlled efficacy study was conducted and published by Polack et al. (2020). This trial followed a 1:1 randomization ratio, assigning over 34,000 participants to either the vaccine or placebo groups. The study was designed to reach statistical maturity upon the occurrence of 170 laboratory-confirmed COVID-19 cases in participants without prior evidence of infection. The resulting data demonstrated a significant disparity in outcomes. Of the 170 confirmed cases, 8 occurred in the BNT162b2 cohort ($n = 17,411$), while 162 occurred in the placebo cohort ($n = 17,511$). This results in an observed infection rate of approximately 0.046% for the vaccine group compared to 0.925% for the placebo group.

```r
trial_matrix <- matrix(c(8, 17411-8, 162, 17511-162),
                       nrow = 2,
                       dimnames = list(Outcome = c("Infected", "Healthy"),
                                       Group = c("BNT162b2", "Placebo")))


rates <- trial_matrix["Infected", ] / colSums(trial_matrix)
barplot(rates,
        main="Infection Rate by Group",
        ylab="Proportion of Cases",
        col=c("skyblue", "salmon"),
        ylim=c(0, 0.01))
```




```
array <- as.array(trial_matrix)
mosaicplot(array, main="Mosaic Plot of Distribution", sub = NULL, xlab = NULL, ylab = NULL,
           sort = NULL, off = NULL, dir = NULL,
           color = c("skyblue","salmon"), shade = FALSE, margin = NULL,
           cex.axis = 0.66, las = par("las"), border = NULL)
```



This analysis seeks to evaluate the efficacy of the vaccine based on these findings. Specifically, we test whether the vaccine efficacy (ψ) is significantly greater than the 30% threshold required for regulatory approval. We formulate our research goal through the hypotheses $H_0 : \psi = 0.3$ versus $H_1 : \psi > 0.3$.

Statistical Methods

Model

Describe the statistical model used.

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

The Bayesian approach requires the specification of a prior distribution for π that represents our state of knowledge before observing the trial data. We adopt a conjugate Beta prior, $\pi \sim Beta(a, b)$, which is mathematically advantageous as it ensures the posterior distribution will also belong to the Beta family.To determine the values of the hyperparameters a and b, we perform a prior elicitation based on a "pessimistic" perspective of vaccine performance. We assume a prior median for π of 0.5, reflecting an initial belief that the vaccine and placebo groups are equally likely to result in an infection. Furthermore, we reflect regulatory skepticism by setting only a 5% prior probability that the vaccine efficacy meets the 30% threshold ($P(\psi \geq 0.30) = 0.05$). Converting this efficacy threshold to the π scale using the transformation $\pi = \frac{1-\psi}{2-\psi}$, we target a distribution where the 5th percentile is approximately 0.4118. Solving these constraints yields the parameters:

$$a = 43.04, \quad b = 43.04$$

The likelihood of observing t infections in the vaccine group out of n total cases is given by the binomial probability mass function:

$$f(t|\pi) = \binom{n}{t} \pi^t (1-\pi)^{n-t}$$

Applying Bayes' Theorem, the posterior distribution $f(\pi|t)$ is proportional to the product of the likelihood and the prior. Due to the conjugacy of the Beta-binomial model, the posterior distribution for π is updated by adding the observed successes and failures to the prior parameters:

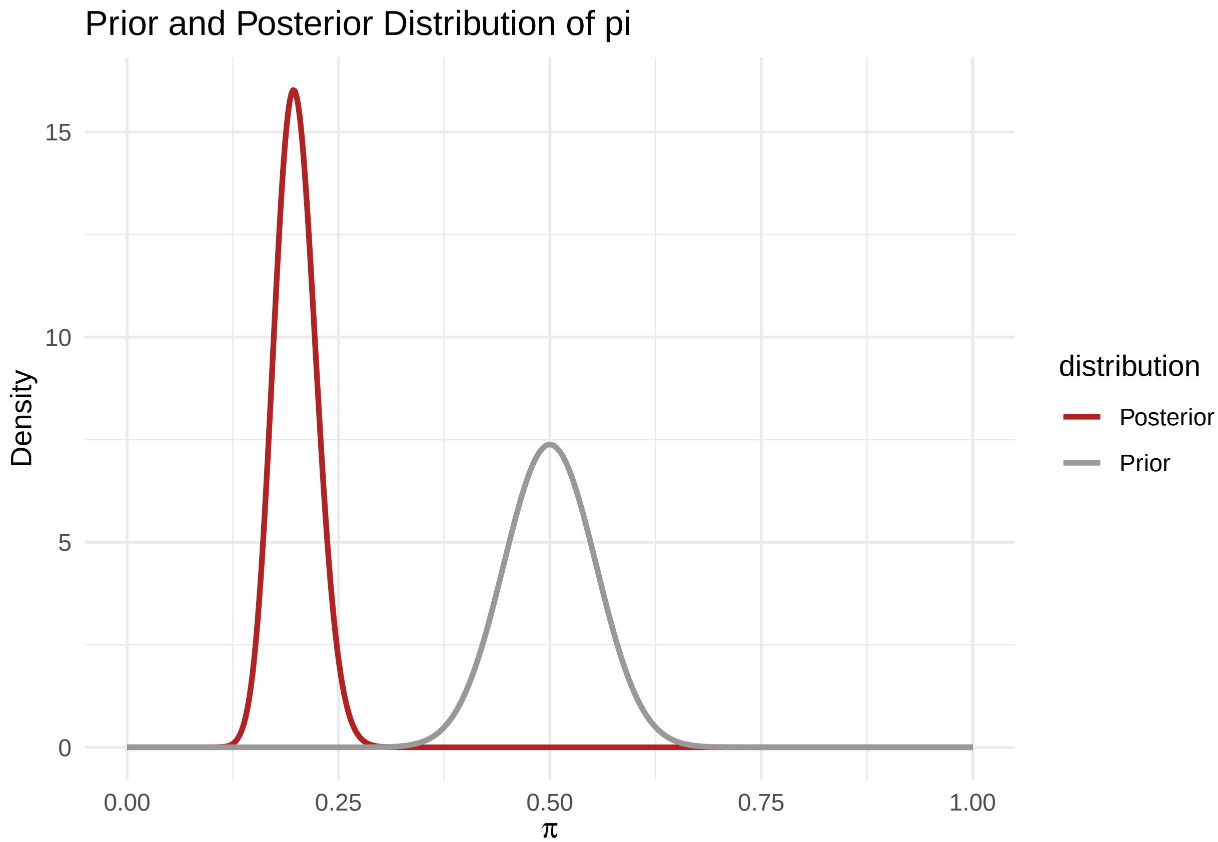
$$\pi|t \sim Beta(a + t, b + n - t)$$

With our observed data of $t = 8$ and $n = 170$, the posterior becomes $\pi|t \sim Beta(51.04, 205.04)$. Following the calculation of this distribution, we transform the results back to the ψ scale. This allows for the determination of the posterior median for vaccine efficacy and a 95% Highest Posterior Density Interval (HPDI) to quantify uncertainty. Finally, the posterior probability $P(\psi > 0.30|t)$ is calculated to evaluate the strength of evidence in favor of the research hypothesis.

```r
# Define parameters
a_prior <- 43.04
b_prior <- 43.04
t_obs <- 8
n_total <- 170
a_post <- a_prior + t_obs
b_post <- b_prior + (n_total - t_obs)

# Generate data for plotting
x <- seq(0, 1, length.out = 500)
df <- data.frame(
  pi = rep(x, 2),
  pdf = c(dbeta(x, a_prior, b_prior), dbeta(x, a_post, b_post)),
  distribution = rep(c("Prior", "Posterior"), each = 500)
)

# Plot
ggplot(df, aes(x = pi, y = pdf, color = distribution)) +
  geom_line(linewidth = 1) +
  labs(title = "Prior and Posterior Distribution of pi",
       x = expression(pi),
       y = "Density") +
  theme_minimal() +
  scale_color_manual(values = c("Prior" = "gray60", "Posterior" = "firebrick"))
```



Results

The Bayesian analysis provides a probabilistic assessment of the vaccine efficacy parameter, ψ, derived from the posterior distribution of π. Using the pessimistic prior distribution ($Beta(43.04, 43.04)$) and the trial data of 8 vaccine infections out of 170 total cases, the posterior distribution for π was found to be $Beta(51.04, 205.04)$.The posterior median for the probability that an infected individual belonged to the vaccine group, π, is 0.199. When transformed to the vaccine efficacy scale using the relationship $\psi = \frac{1-2\pi}{1-\pi}$, this yields a posterior median vaccine efficacy of 75.1%. The uncertainty in this estimate is characterized by a 95% Highest Posterior Density Interval (HPDI) for ψ of [66.5%, 82.2%].To evaluate the research hypothesis $H_1 : \psi > 0.3$, the posterior probability was calculated. The analysis indicates that the posterior probability of the vaccine efficacy exceeding the 30% regulatory threshold, $P(\psi > 0.30|\text{data})$, is greater than 0.9999. These inferential statistics are summarized in Table 2.

Table 2: Bayesian Posterior Summaries for Vaccine Efficacy (ψ)| Metric | Posterior Estimate || :— | :— || Posterior Median (ψ) | 75.1% || 95% HPDI (ψ) | [66.5%, 82.2%] || $P(\psi > 0.30 \mid \text{data})$ | > 0.9999 |

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Grove, R. (2026). *Physicians-Health-Study.pdf.* Lecture slides, STAT 342, University of Washington.

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Polack, F. P., Thomas, S. J., Kitchin, N., Absalon, J., Gurtman, A., Lockhart, S., . . . Gruber, W. C. (2020). Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine. New England Journal of Medicine, 383(27), 2603-2615.

Appendix

Code

```
# You can reference your code in the appendix (sample here).
```

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.